

Zihan (Tomson) Li

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EDUCATION

Washington University in St. Louis

Doctor of Philosophy in Computer Science

St. Louis, MO

Aug. 2023 – Present (Expected May 2028)

Washington University in St. Louis

Master of Science in Cybersecurity Engineering

St. Louis, MO

Aug. 2022 – May 2023

Washington University in St. Louis

Bachelor of Science in Computer Engineering

St. Louis, MO

Aug. 2020 – Dec 2022

DePauw University

Bachelor of Arts

Greencastle, IN

Aug. 2017 – May 2020

EXPERIENCE

Graduate Research Assistant

Washington University in St. Louis

May 2022 – Present

St. Louis, MO

- Research focuses on **system security and cyber-physical systems (CPS)**.
- Analyzed IoT firmware updates, examining 150 images from 33 device families. Discovered **zero-day and n-day** vulnerabilities, leading to **25 CVE IDs and one PSV ID** through responsible disclosure.
- Developed experimental Linux scheduler enforcer for **timing violation detection and mitigation** with an **average performance overhead of only around 2.8%**.
- Optimization on communication cost of Federated Learning. Reducing communication cost by **30%** while maintaining training accuracy of **95%**.
- Discovery of timing impact on security protection mechanisms for CPS. Developed an optimization framework that **mitigates attacks while maintaining schedulability for real-time CPS**.

C++ Backend Development Intern

Youme.im

July 2020 – Sept. 2020

Shenzhen, China

- Developed an **end-to-end encryption module** for audio and video real-time communication
- Integrated the encryption module into the cross-platform compilation build workflow
- Utilizing optimization techniques to ensure **low-performance overhead** for encryption and decryption.

PUBLICATIONS

Conference Papers

Balancing Security and Schedulability: WCET Evaluation and Security Optimization in CPS

2026 IEEE 32nd Real-Time and Embedded Technology and Applications Symposium (RTAS)

Zihan Li, Marion Sudvarg, Ching-Hsiang Chan, Ryan Burrow, Nathan Burow, Cailani Lemieux-Mack, Sanjoy Baruah, Ning Zhang, Bryan C. Ward

Sensing and Stopping Interfering Secondary Users: Validation of an Efficient Spectrum Sharing System

2026 IEEE International Symposium on Spectrum Innovation (DySPAN)

Meles G Weldegebriel, **Zihan Li**, Dustin Maas, Gregory Hellbourg, Ning Zhang, Neal Patwari

Resilient Federated Learning on Embedded Devices with Constrained Network Connectivity

2025 62nd ACM/IEEE Design Automation Conference (DAC)

Zihan Li, Han Liu, Ao Li, Ching-Hsiang Chan, Yevgeniy Vorobeychik, William Yeoh, Wenjing Lou, Ning Zhang

A Unified Hardware Performance Profiling Infrastructure to Measure and Manage Uncertainty

19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)

Ao Li, Marion Sudvarg, **Zihan Li**, Sanjoy Baruah, Chris Gill, Ning Zhang

Your Firmware Has Arrived: A Study of Firmware Update Vulnerabilities

33rd USENIX Security Symposium (USENIX Security 24)

Yuhao Wu, Jinwen Wang, Yujie Wang, Shixuan Zhai, **Zihan Li**, Yi He, Kun Sun, Qi Li, Ning Zhang

Conference / Workshop Short Papers

Spectrogram-Based Interference Source Attribution at OVRO: An Over-the-Air Validation Using Pseudonymetry

To Appear in 2026 IEEE RFID NRDZ Workshop

Meles G Weldegebriel, [Zihan Li](#), Gregory Hellbourg, Ning Zhang, Neal Patwari

Tintin: PMU Scheduling to Minimize Uncertainty

OSPERT Workshop at ECRTS 2025

Marion Sudvarg, Ao Li, [Zihan Li](#), Sanjoy Baruah, Chris Gill, Ning Zhang

Work-in-Progress: Measuring Security Protection in Real-time Embedded Firmware

2022 IEEE Real-Time Systems Symposium (RTSS)

Yuhao Wu, Yujie Wang, Shixuan Zhai, [Zihan Li](#), Ao Li, Jinwen Wang, Ning Zhang

PROJECTS

Open Platform for Cyber Physical System Research (OPCPS)

April 2025 – Present

- Design and implement an open-source platform specifically tailored for CPS security research testing
- Offers modular components designed for trivial replacement with alternative implementations.
- Security instrumentations for various real-world CPS platforms
- Performance and timing profiling to understand the real-time impact of CPS under different security instrumentations.

EMILY: Electro Magnetic Interference Ledger & registrY | *EMI, Database*

Nov 2025 – Present

- A database to provide interference insights for radio astronomy.
- Design database structure to improve the performance and maintainability of EMILY.
- Implement backend for high performance and secure access to the database based on Springboot framework.
- Design and implement a user-friendly front-end for astronomers to use the system with ease.
- Real-world impact and public available at emilyknows.info

Federated Learning Optimization | *Machine Learning, Federated Learning, Python*

Aug 2024 – Dec 2024

- Designed adaptive ML protocols for distributed predictive modeling, using **PyTorch, pandas, and NumPy**. Cut communication costs by **30%** while preserving **95%** training accuracy.
- Conduct an empirical study on communication cost for federated learning.
- Designed and optimized data pipelines for processing large-scale distributed datasets. Adaptive gradient compression rate can reach up to **210x**
- Real-world FL simulation demonstrates the feasibility of the proposed approach.
- Accepted by *2025 62th ACM/IEEE Design Automation Conference (DAC)*

HONOR AWARDS

2022 Dean's Select PhD Fellowship | *Washington University in St. Louis*

Nominated for the 2022 Dean's Select PhD Fellowship at Washington University in St. Louis.

Dean's List | *DePauw University*

Recognized on the Dean's List for 2017 and 2020

SERVICES

Reviewer

- IEEE Transactions on Information Forensics and Security
- IEEE/ACM Transactions on Networking
- ACM Transactions on Cyber-Physical Systems
- ISOC Symposium on Vehicle Security and Privacy (VehicleSec '24)
- International Conference on Computer Communications and Networks (ICCCN)

Artifact Evaluation Program Committee

- ACM Conference on Computer and Communications Security (CCS)

TEACHING

CSE 5619: Recent Advances in Computer Security and Privacy

Spring 2026

- AI (Assistant in Instruction) and class coordinator

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, HTML/CSS, VHDL, Assembly, SQL, PHP

Frameworks: React, Node.js, ROS, ROS2

Developer Tools: Git, Cmake, Docker, VS Code, Visual Studio, Eclipse, Wireshark, Xcode, Ghidra, Database Management Systems, Excel, Gazebo

Libraries: pandas, NumPy, Matplotlib, Tkinter, Pytorch

OS: Linux Kernel Programming, Kernel Scheduler, Kernel Network Stack